

XiloVR: Simulation for Teaching and Technique Preservation

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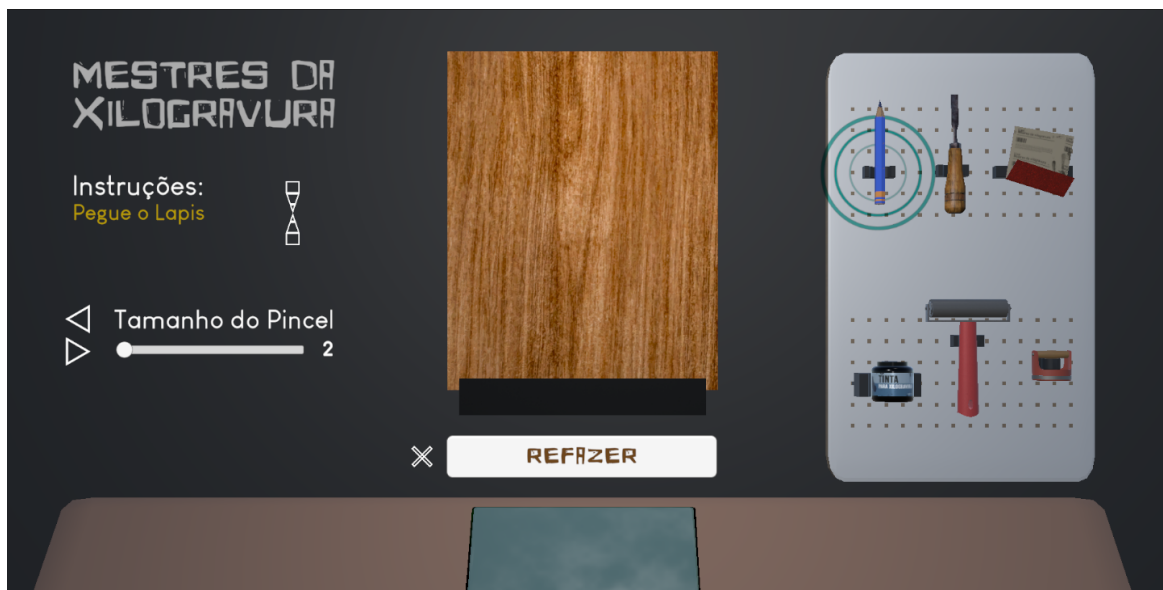


Figure 1: Environment capture of the experience, 2025.

ABSTRACT

This demonstration presents a new approach to exploring the art of woodblock printing through an interactive Virtual Reality (VR) environment, leveraging the educational potential of immersive technologies to promote art and culture. In this context, a physical prototype was designed to replace the manual interactions typically performed with VR controllers. Instead, the system uses sensor-tracked representations of woodcut tools to enable interaction in the immersive environment. Its goal is to offer a practical, educational and immersive experience in artistic creation, innovating teaching methods and preserving the knowledge of woodcut printing for future generations. By combining physical and digital elements, this solution bridges technological accessibility with cultural heritage promotion, offering an engaging and transformative learning platform.

Index Terms: Xylography, Virtual Reality, Art teaching

1 INTRODUCTION

Woodcut printing is a printmaking technique that uses wood as a matrix to communicate visual narratives, ideas, and traditions across various cultures worldwide [1]. Simulations, in turn, are

interactive digital learning environments that mimic real-life processes or situations ([4], [3], and [7] apud [5]). These simulations enable learners to practice skills that might otherwise be dangerous in real-life scenarios within a safe environment. Furthermore, they provide cost-effective alternatives for practicing procedures using virtual devices that might be prohibitively expensive in reality ([5] and [2]).

In this context, XiloVR emerges as a tool to disseminate and enhance the learning of woodcut techniques through unlimited resources and immersive practices provided by digital simulation in virtual reality. In addition, it integrates augmented reality to simulate the step-by-step process of the real technique using physical tools tracked in the virtual environment. A customized support structure was also developed to replace the need for VR headsets, thus increasing the accessibility of the project. This article presents the initial motivations behind XiloVR's innovative approach, emphasizing the determining factors that shaped its development and providing a detailed description of the system's overall architecture.

2 APPLICATION

2.1 Physical and Virtual Resources

The immersive experience was developed using the Unity 2022.3 game engine, with XR development packages such as the XR Interaction Toolkit and XR Plugin Management. The Vuforia package for tracking flat images and triggering interactive events was also integrated. A virtual environment was created to simulate an artistic production studio in a realistic way, with 3D models developed in Blender 4.0.

The project was initially designed for the Quest 2 headset, but to make the experience more accessible, a cost-effective prototype

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was developed using a cardboard and plastic box. This prototype supported a webcam that tracked figures representing woodcut tools, which were attached to cardboard cards with wooden supports for stability. This solution allowed for broader participation without relying on advanced VR headsets.

2.2 Immersive Experience

To initiate the experience, participants positioned themselves in front of the cardboard box, which housed the webcam. The plastic-covered side of the box faced the participant, allowing the camera to capture their movements. The immersive environment was rendered and displayed on a monitor for real-time interaction.

At the onset, the system provided contextual information about the environment and visual representations of the figures associated with each stage of the woodcut production process. Participants could choose between selecting a pre-designed artwork by interacting with the figure of two arrows in opposite directions or initiating a new simulation to create an artwork manually by selecting the figure of an X.

A blue guiding light and an instruction bar supported participants step by step in the virtual woodcut process. They began by selecting the pencil tool to draw on the virtual wood surface, followed by the gouge for carving, sandpaper for refinement, and ink with a roller for coloring. A virtual sheet of paper was then placed on the woodcut, and the barren tool was used to press the paper. After finalizing with the arrow tool, the completed artwork was revealed and could be exported to the cloud for printing and preservation as a keepsake.

2.3 Architecture

The system used the Vuforia database to track figures presented to the camera. Images of the physical figures were stored and synchronized in Unity with their corresponding virtual tools. When a figure was detected, the system queried the database and synchronized the image with its associated 3D model.

In the virtual environment, the core simulation mechanics centered on using virtual tools to modify the textures and appearance of the woodcut matrix's 3D model. To achieve this, a technique inspired by scratch-off lottery tickets was implemented, enabling participants to reveal modifications incrementally. Two shaders were developed: the management shader, which handled texture transitions during each stage, and the scratch shader, which revealed textures interactively as participants engaged with the tools.

Texture management was implemented using Unity's Shader Graph, leveraging the Lerp function to blend two input textures based on a grayscale mask. Each stage of the simulation (e.g., drawing, carving, and painting) incorporated specific texture layers to produce the desired visual effects. A critical design feature ensured that areas drawn during the initial stage remained unaltered during subsequent carving stages, preserving the integrity of the artwork and enhancing the user experience.

Interactive functionality was achieved by emitting a ray from the participant's virtual tool (e.g., a pencil), detecting collision points on the object, and mapping the detected coordinates to texture data. This allowed real-time visual updates on the woodcut matrix. This architectural framework provided an efficient mechanism for texture manipulation, enabling a seamless simulation of traditional woodcut techniques.

3 MOTIVATION AND INNOVATION

The characteristics of virtual environments, such as active presentation of materials and unlimited access to learning resources, have proven to be more effective in enhancing knowledge and work-related skills ([6] apud [5]). Accordingly, XiloVR is motivated by the desire to enhance the learning process of woodcut printing techniques and their multiple applications in artistic and literary fields,

as well as the ambition to preserve this ancient technique, which has significantly contributed to the documentation of art, narratives, and traditions of various cultures worldwide.

Furthermore, meta-analyses conducted by Sitzmann (2011) and Vogel et al. (2006) on the effects of computer-based interactive games and simulations reveal that students performed better when they controlled their navigation within the virtual learning environment compared to scenarios where the teacher controlled the learning process ([8] and [6] apud [5]). According to Lee's meta-analysis, simulations are effective both for presenting and practicing when combined with other instructional methods. His findings also indicate that students performed better when some form of guidance was provided, even during practice sessions, compared to scenarios without guidance ([4], [8], and [6] apud [5]).

In XiloVR, users take control of the artistic process, independently navigating through all stages of creating artwork. Additionally, they receive both visual and textual instructions at each step to ensure a clear understanding of the progression, thereby optimizing the learning experience. Moreover, with the inclusion of an Augmented Reality (AR) version, the system offers an immersive and more accessible experience, aiming to reach as wide an audience as possible. Thus, this research seeks to evaluate the effectiveness of Augmented Reality in preserving the woodcut printing technique, exploring its use in teaching, documenting, and disseminating this traditional art form.

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